Lecturers: Thomas Quella and Nick Beaton
School of Mathematics and Statistics

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University of Melbourne

MAST20030: Differential Equations
Tutorial: Week 5

Question 1

Do the following improper integrals converge or diverge?

(a) $\displaystyle\int_0^\infty \frac{1}{1+t^2}\,\mathrm{d}t.$

(b) $\displaystyle\int_0^\infty \frac{t}{1+t^2}\,\mathrm{d}t.$

(c) $\displaystyle\int_0^\infty \mathrm{e}^{-t}\cos(\mathrm{e}^{t^2})\,\mathrm{d}t.$

Question 2

(a) Using the integral definition, find the Laplace transform of t^2. For what values of s is the Laplace transform defined?

(b) Find the Laplace transform of

$$f(t) = 3\mathrm{e}^{-4t} + \sinh\frac{t}{2} + \frac{t^5}{2}$$

Question 3

Write down the Laplace transform, if it exists, of the following functions:

(a) $(t+7)^2\cosh t.$

(b) $\mathrm{e}^{(t-1)^2}.$

(c) $\cos(\omega t+\theta)$, where ω and θ are constants.

State the values of s for which the Laplace transform is defined.

Question 4

Find the inverse Laplace transforms of the following functions:

(a) $\dfrac{2s+7}{s^2+16}.$

(b) $\dfrac{12}{s^2-2s-3}.$

(c) $\dfrac{s^2+9s-2}{(s^2+16)(s-2)}.$

(d) $\dfrac{5}{(s-3)^4}.$

(e) $\dfrac{4s^2-s+10}{s^2\,(s+1)}$

(f) $\dfrac{s}{(s+4)^4}.$